INTELLIGENT ELEPHANT CARE AND VISITOR INSIGHTS: A MULTI-COMPONENT RESEARCH STUDY AT PINNAWALA ELEPHANT ORPHANAGE

Project ID: 25-26J-391

Final Report Draft

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PERSONALIZED ELEPHANT FOOD CHAIN RECOMMENDATION

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Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Elephants are among the most iconic species in Asia and play a vital ecological, cultural, and economic role. However, improper dietary management, overfeeding, and nutrient deficiencies often lead to health issues, reduced lifespan, and higher maintenance costs in captive elephants. Traditional feeding practices mostly rely on generalized estimates rather than the specific nutritional requirements of each animal, overlooking critical factors such as age, body condition, activity level, and environmental conditions.

This research focuses on developing a Personalized Elephant Food Chain Recommendation System that leverages data-driven approaches to optimize nutrition for individual elephants. The study integrates multi-source data, including elephant health profiles, activity monitoring, and environmental parameters, to generate precise dietary recommendations. A combination of rulebased methods, nutritional requirement models, and machine learning techniques will be employed to formulate optimal feeding plans while ensuring animal welfare.

The system aims to minimize health risks associated with malnutrition, enhance feeding efficiency, and reduce operational costs for elephant care facilities. Furthermore, the proposed solution supports conservation and wildlife management by introducing technology-driven precision nutrition for elephants in Sri Lanka and beyond.

The outcomes of this research are expected to contribute to both academic knowledge and practical applications, offering a scalable framework that can be extended to other large herbivores in captive and semi-wild environments.

Keywords: Elephant Nutrition, Personalized Feeding, Food Chain Recommendation, Machine Learning, Animal Welfare, Conservation.

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1. INTRODUCTION

Elephants are one of the most significant terrestrial mammals, both ecologically and culturally, particularly in Asian countries such as Sri Lanka. They play a critical role as ecosystem engineers by influencing vegetation structure, seed dispersal, and habitat dynamics. Beyond their ecological importance, elephants also hold strong cultural and religious significance, making their conservation and welfare an essential responsibility. However, managing elephants in captivity or semi-captive environments remains a major challenge, especially in terms of nutrition and overall health management.

Feeding practices for elephants are often based on generalized guidelines rather than individual needs. Such practices may lead to overfeeding, nutrient imbalances, or deficiencies, resulting in obesity, metabolic disorders, or weakened immunity. Factors such as age, weight, reproductive status, health conditions, activity levels, and seasonal forage availability are rarely considered when formulating diets. This lack of personalization highlights the need for a systematic approach to ensure that each elephant receives an optimal diet tailored to its unique requirements.

Recent advances in data science, machine learning, and IoT-enabled monitoring provide new opportunities to improve animal nutrition and welfare. By integrating health records, environmental data, and activity tracking, it becomes possible to build personalized dietary recommendation systems that address the limitations of traditional feeding practices. Such systems can support veterinarians, caretakers, and wildlife managers in making informed decisions regarding elephant diets, ultimately enhancing both animal well-being and cost efficiency in elephant care facilities.

This research proposes a Personalized Elephant Food Chain Recommendation System that generates data-driven feeding plans for individual elephants. The system is designed to incorporate multiple factors, including health metrics, environmental conditions, and activity levels, to produce optimized nutritional recommendations. The outcomes of this study are expected to contribute to improving elephant welfare, reducing the risks of malnutrition, and supporting sustainable elephant management practices.

1.1. BACKGROUND & LITERATURE SURVEY

Elephants have complex dietary needs that vary with age, sex, health, and activity level. In the wild, they consume diverse vegetation, but captive elephants often rely on standard rations that may cause malnutrition, obesity, or metabolic issues.

Research shows that monitoring body condition, activity, and nutrient intake is essential for optimal health. Modern technologies, including IoT sensors and machine learning, enable real-time tracking of elephant activity and environment, supporting data-driven feeding decisions.

Most existing feeding systems are rule-based and do not account for individual variation. Studies in livestock demonstrate the benefits of personalized, AI-driven diets, yet research on elephant-specific personalized nutrition—especially in Sri Lanka—is limited.

This highlights the need for a Personalized Elephant Food Chain Recommendation System, which integrates health, activity, and environmental data to optimize nutrition, improve welfare, and reduce feed waste.

1.2.RESEARCH GAP

Elephants have complex dietary requirements that vary with age, sex, health status, reproductive condition, and activity levels. In the wild, elephants select diverse vegetation that naturally balances their nutrient intake. However, in captive or semicaptive environments, feeding practices largely rely on generalized feeding guidelines or standard rations. Such approaches fail to address individual variations, often leading to malnutrition, obesity, metabolic disorders, or weakened immunity, which adversely affect elephant welfare and longevity.

Several studies emphasize the importance of personalized nutrition in large mammals, particularly in zoo settings where body condition scores (BCS), activity levels, and nutrient intake monitoring have been shown to improve health outcomes. Modern technologies such as IoT-based activity tracking, wearable sensors, and machine learning models have enabled real-time monitoring of animal behavior, health metrics, and environmental factors, providing opportunities for data-driven dietary management.

Despite these advances, research specifically targeting personalized feeding systems for elephants remains limited, especially in the Sri Lankan context. Most existing models are either rule-based, following fixed nutritional guidelines, or are designed for livestock, with minimal adaptability for elephant physiology and behavior. Additionally, there is a lack of integration between health records, environmental conditions, and activity data, which are critical for formulating optimized feeding plans.

This clear gap in the literature demonstrates the necessity for a comprehensive, datadriven, and elephant-specific dietary recommendation system. Such a system would not only provide personalized nutrition plans tailored to each elephant's unique profile but also support efficient resource use, reduce feed wastage, and enhance overall animal welfare. By addressing these gaps, this research aims to contribute both to academic knowledge and practical applications in elephant conservation and management.

1.3. summary table about the research gap.

Table 1Summary Table About The reseach gap

Area	What exists in prior work	Gap for semi-tamed elephants	Why it matters
Nutritional Guidelines	Standard feeding guidelines exist for captive elephants; studies focus on general nutrient requirements and body condition scoring	Guidelines rarely account for individual variation in age, health, activity level, or environmental conditions for semitamed elephants.	Without personalized diets, elephants may face malnutrition, obesity, or metabolic disorders, reducing welfare and lifespan
Activity & Behavior Monitoring	Wearable sensors and IoT devices have been used in livestock and zoo elephants to track movement and activity.	Semi-tamed elephants in Sri Lanka are less monitored, and integration of activity data with diet planning is minimal.	Lack of activityinformed feeding leads to inaccurate energy intake, affecting health and energy balance.
Environmental Factors	Some studies incorporate forage availability and seasonal changes in diet recommendations for wildlife and livestock.	For semi-tamed elephants, local environmental factors and seasonal forage variability are rarely included in diet planning.	Ignoring environmentdriven dietary needs can cause nutrient deficiencies or overfeeding, increasing feed costs.

Data-Driven / AI Approaches	ML and optimization models are used in livestock for personalized diets and health prediction.	Few studies apply machine learning or optimization for semitamed elephant nutrition.	Using AI can improve feeding efficiency, reduce feed waste, and enhance individual welfare.

1.4. RESEARCH PROBLEM

Captive and semi-tamed elephants often suffer from nutrition-related health issues due to generalized feeding practices that do not consider individual differences. Factors such as age, sex, body condition, reproductive status, activity level, and environmental conditions significantly influence their dietary requirements. However, most existing feeding systems rely on standard guidelines and fail to integrate real-time data on activity, health metrics, or environmental changes.

This leads to several problems:

1. Malnutrition or overfeeding, causing obesity, metabolic disorders, and weakened immunity.
2. Inefficient feed utilization, resulting in higher costs and resource wastage.
3. Limited applicability of current research on semi-tamed elephants, particularly in the Sri Lankan context, where environmental and management conditions differ from zoo or fully wild settings.

Despite advancements in IoT, wearable sensors, and AI-driven dietary management in livestock and zoo animals, there is no comprehensive system that provides personalized, data-driven feeding recommendations for semi-tamed elephants. This research aims to address this gap by developing a Personalized Elephant Food Chain Recommendation System that optimizes nutrition based on individual profiles, activity data, and environmental conditions, ultimately improving elephant welfare and feeding efficiency.

2. RESEARCH OBJECTIVES

2.1.MAIN OBJECTIVE

The main objective of this research is to develop a Personalized Elephant Food Chain Recommendation System that provides data-driven, individualized dietary plans for semi-tamed elephants. The system aims to integrate health metrics, activity levels, and environmental conditions to optimize nutrition, enhance animal welfare, and improve feeding efficiency.

2.2. SPECIFIC OBJECTIVES

2.2.1 user requirements

- Provide individualized dietary recommendations based on elephant specific data (age, weight, BCS, health status, and activity).
- Integrate environmental and seasonal factors affecting forage availability and nutrient content.
- Offer easy-to-use interfaces for data input, visualization, and recommendation reports.
- Generate alerts and notifications for feeding schedules, nutritional deficiencies, or health concerns.
- Support data storage and historical tracking to monitor trends in elephant health and diet effectiveness.

2.2.2 functional requirements Elephant Profile Management

Ability to store and update individual elephant data, including age, sex, weight, body condition score (BCS), health history, and activity levels.

Data Collection & Integration

Integration of data from various sources such as **health records, wearable activity trackers, environmental sensors, and forage availability databases**.

Diet Recommendation Engine

Generate **personalized daily feeding plans** based on elephant-specific metrics, nutritional requirements, and environmental conditions.

Optimize macro- and micronutrient intake to maintain health and prevent overfeeding or deficiencies.

Alert & Notification System

Notify users about **feeding schedules, dietary adjustments, or potential health risks** based on changes in activity, health, or environment.

Data Analysis & Reporting

Provide **visual dashboards** and reports for monitoring trends in elephant health, diet compliance, and nutritional outcomes.

System Usability

Ensure **user-friendly interface** for caretakers and veterinarians with minimal training.

Support **multi-platform access** (desktop and mobile).

Historical Data Tracking

Maintain historical records of **feeding plans, diet adjustments, and health outcomes** for longitudinal analysis and research purposes.

2.2.3 non-functional requirements

Performance: Generate recommendations quickly and handle multiple elephant profiles.
Scalability: Support new elephants or additional data sources without redesign.
Reliability & Availability: Operate continuously with minimal downtime.
Usability: Easy-to-use interface for caretakers and veterinarians.
Security: Protect sensitive elephant health and facility data.
Maintainability: Allow easy updates to algorithms or dietary rules.
Portability: Accessible on both desktop and mobile devices.

3. METHODOLOGY

3.1 Research Approach

This research follows a **Design Science Research (DSR)** methodology, focusing on creating an intelligent system to optimize elephant nutrition. The approach integrates hardware-based data acquisition with a multi-layered AI inference engine to provide actionable feeding recommendations.

3.2 System Architecture Overview

The system is designed with a three-tier architecture, ensuring seamless data flow from the physical environment to the end-user:

Tier	Technology	Role
Data Acquisition (IoT)	GPS, Accelerometer, Acoustic Sensors	Captures real-time physical and vocal data from elephants.
Backend (Inference)	Flask (Python), TensorFlow, Scikit-learn	Processes sensor data and runs the Recommendation Engine.
Interface	React Native / Flutter	Dashboard for Mahouts to receive dietary instructions.

3.3 Module 1 — Text-Based Dietary Log Analysis

Approach: Natural Language Processing (NLP)

- **Step 1:** Ingests daily logs written by mahouts regarding elephant appetite and fodder rejection.
- **Step 2:** Uses **Tokenization** and **Lemmatization** to extract keywords (e.g., "rejected dry grass", "high water intake").

- **Step 3:** A **Sentiment Analysis** model scores the health status (Positive/Neutral/Negative) based on the logs.
- **Step 4:** Outputs a "Dietary Health Score" to the fusion engine.

3.4 Module 2 — Acoustic Stress & Hunger Detection

Approach: Audio Signal Processing + Deep Learning (CNN-GRU)

- **Step 1:** Captures rumbles and trumpets via MEMS microphones on the collar.
- **Step 2:** Converts raw audio into **Mel-Frequency Cepstral Coefficients (MFCCs)** and spectrograms.
- **Step 3:** A **CNN-GRU Hybrid Model** (similar to the CricAI architecture) processes the temporal features to classify the sound into: *Hunger, Stress, or Contentment*.
- **Step 4:** Sends the classified emotional state as a numerical weight to the central engine.

3.5 Module 3 — Mobility & Foraging Analysis

Approach: GPS Trajectory Mapping + Social Inference

- **Step 1:** Collects GPS coordinates and Accelerometer data.
- **Step 2:** Calculates **Movement Entropy** (diversity of locations visited) and **Velocity**.
- **Step 3:** Uses **Geofencing** against Satellite Vegetation Data (NDVI) to see if the elephant is in a high-nutrient zone.
- **Step 4:** Detects "Social Isolation" if the elephant remains distant from the herd's centroid for extended periods.

3.6 Module 4 — The Recommendation Engine (Fusion Model)

Approach: Weighted Decision-Level Fusion

- **Step 1:** The engine receives scores from Text, Audio, and GPS modules.
- **Step 2: Weighted Calculation:**

  $$Final\_Recommendation = (w1 \times NLP\_Score) + (w2 \times Audio\_State) + (w3 \times Mobility\_Index)$$

- **Step 3:** If the score falls below a threshold, the engine triggers a "Dietary Shift" alert (e.g., "Substitute 20% Coconut Palm with fresh Kithul").

3.7 Hardware Integration Architecture

The hardware components are integrated into a **Smart Collar Unit**:

- **Microcontroller:** Arduino Nano 33 BLE (Edge processing).
- **Sensors:** MPU6050 (Activity), NEO-6M (GPS), High-sensitivity Mic (Audio).
- **Connectivity:** LoRa/SIM800L for long-range data transmission to the Flask server.

3.8 Development Phases

1. **Phase 1 — Data Collection:** Gathering historical feeding logs and recording vocalizations at Pinnawala.
2. **Phase 2 — Feature Engineering:** Extracting MFCCs from audio and cleaning GPS traces.
3. **Phase 3 — Model Training:** Training the CNN-GRU and NLP classifiers.
4. **Phase 4 — System Integration:** Fusing all modules into the Flask backend.
5. **Phase 5 — Validation:** Comparing AI recommendations with veterinary expert logs.

3.1.2 Software Component

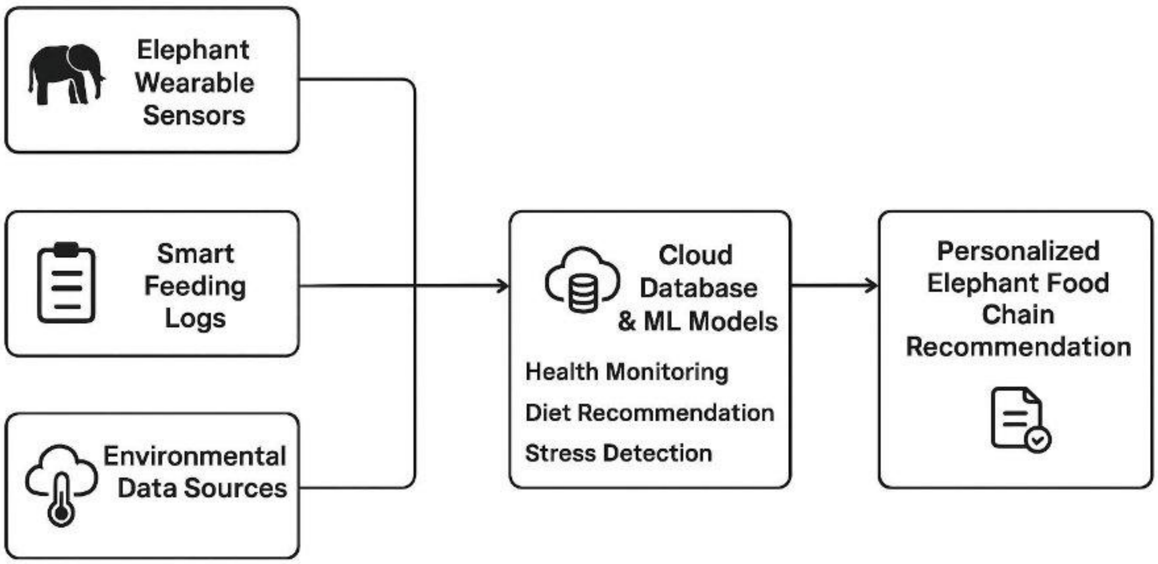

Figure 2 Software Component

4. **Description of Personal and Facilities**

4.1 hardware component

- Microcontroller/Processor: Controls the system and processes data (e.g., Arduino, Raspberry Pi).
- Sensors: Collect data such as temperature, motion, or location.
- Actuators/Output Devices: Perform actions like turning on motors, LEDs, or alarms.
- Power Supply: Provides energy to all components.
- Communication Modules: Enable data transfer (Wi-Fi, Bluetooth).
- Peripherals: Displays or buttons for user interaction**.**

4.2. System Technology

- **Programming Languages:** Python, Java, or C/C++ for system logic and data processing.
- **Database:** MySQL, SQLite, or Firebase for storing collected data.
- **IoT & Communication:** Wi-Fi, Bluetooth, or GSM modules for real-time data transfer.
- **Cloud & Web Services:** Cloud platforms for data storage, analytics, and remote monitoring.
- **User Interface:** Web or mobile applications for visualization and control.

4.3 Programming Languages

- **Python:** For data processing, sensor integration, and backend logic.
- **Java:** For creating user interfaces and mobile applications (if required).
- **C/C++:** For programming microcontrollers and hardware-level control.
- **HTML/CSS/JavaScript:** For web-based dashboards and data visualization.

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